

Soham Lodh

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SUMMARY

Computer Science undergraduate (CGPA: 9.31) with hands-on experience architecting concurrent, distributed-style systems: event-driven pipelines, atomic score aggregation under concurrent load, and multi-tenant SaaS backends enforcing consistency at the data layer. Built and shipped production systems end-to-end in JavaScript, TypeScript, applying data structures, algorithm design, and software engineering principles to solve real concurrency and consistency problems.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Java, C, Python

Frontend: React.js, Next.js, Redux Toolkit, Vite, Tailwind CSS

Backend: Node.js, Express.js, REST API Design, JWT, Clerk Auth, FastAPI, Prisma ORM

Databases: PostgreSQL, MongoDB, Neon Serverless

Cloud & DevOps: Vercel, Render, CI/CD Deployment

Tools & Platforms: Git, GitHub, Postman, Cloudinary

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

ML & Data: NumPy, Pandas, Scikit-learn, XGBoost, SHAP

EXPERIENCE

Technical Member – Web Development

Aug 2025 – Present

Coding Ninjas KIIT Chapter

Bhubaneswar, India

- Architected a distributed-style event management platform (Next.js, Hono, Prisma, MongoDB) featuring a real-time leaderboard that aggregated concurrent judge and audience scores; engineered per-team voting windows, one-vote-per-user consistency guarantees, score normalization, and atomic aggregation to eliminate race conditions during high-concurrency updates.
- Built a secure, server-side quiz platform with session tracking, tab-switch detection, timed execution, and randomized question delivery; diagnosed and resolved session edge cases that surfaced only under concurrent user load, reproducing failure conditions from runtime logs.
- Designed a Project Management System (Next.js, NextAuth, MongoDB) enforcing four-tier role-based access control via middleware; authored API contracts and permission-logic documentation consumed by downstream contributors.

Open Source Contributor

Jul 2025 – Oct 2025

GirlScript Summer of Code

Remote

- Implemented four authentication modules – OTP login, JWT session management, Google OAuth, and password recovery – writing unit tests covering success and failure paths for each flow; changes were reviewed, approved, and merged into the production main branch.
- Reproduced production-level defects by reconstructing failure conditions from logs and submitted targeted fixes merged into the codebase, improving system reliability across affected flows.

PROJECTS

Core Work – Multi-Tenant Project Management Platform

Jun 2026

React, Redux Toolkit, Express, Prisma, PostgreSQL (Neon Serverless), Clerk, Inngest, Vercel

Live: core-work.vercel.app *GitHub:* github.com/Soham-Lodh/Core-Work

- Architected a multi-tenant SaaS backend on a normalized 7-model PostgreSQL schema with cascading deletes and unique-constraint guards against duplicate membership; enforced two-tier role-based access control (workspace ADMIN/MEMBER, project team_lead) across 10+ REST endpoints secured behind Clerk JWT middleware.
- Engineered an event-driven sync pipeline using Inngest to process seven Clerk webhook event types (user and organization lifecycle) into durable background functions, propagating workspace and membership state changes to the database with guaranteed delivery and zero manual intervention.
- Implemented optimistic Redux state management to eliminate full-page refetches on task and project mutations, reducing perceived latency for all CRUD operations; built analytics dashboards aggregating task completion rates, status, and priority distributions across workspaces.

RiskLens – Credit Default Risk Prediction System

Jun 2026

Python, Logistic Regression, FastAPI, React, Vercel, Render

Live: credit-default-risklens.vercel.app *GitHub:* github.com/Soham-Lodh/RiskLens

- Designed and implemented an 18-stage ML pipeline over 50,000 records from three relational data sources: applied stratified splitting before EDA to prevent data leakage, engineered three ratio features from raw bureau data, and used Bayesian hyperparameter optimization (50 trials) to achieve AUC-ROC 0.9834 and KS 85.91 on held-out data.

- Deployed inference via FastAPI + Uvicorn on Render with a React frontend on Vercel; integrated SHAP explainability and a credit scoring algorithm mapping default probability to a 300–900 scale with institutional risk classification.

Prescripto – Medical Appointment Platform
MERN Stack, JWT, Cloudinary, Vercel
Live: prescripto-o5lf.vercel.app *GitHub:* github.com/Soham-Lodh/Prescripto

Oct 2025 – Jan 2026

- Shipped a full-stack medical platform across two independently deployed frontends – a patient app and a unified Admin/Doctor panel – backed by a centralized Express REST API on MongoDB Atlas, live on Vercel.
- Designed role-based JWT authentication with separate login flows, protected route middleware, and permission-scoped dashboards for three account types; hardened the API with input validation, bcrypt password hashing, and rate limiting to prevent credential abuse.

EDUCATION

KIIT University	2024 – 2028
<i>B.Tech, Computer Science Engineering (AI & ML)</i>	<i>CGPA: 9.31</i>
<i>Relevant Coursework:</i> Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming	
Aditya Academy Senior Secondary	2024
<i>Class XII (CBSE)</i>	<i>85%</i>
Aditya Academy Senior Secondary	2024
<i>Class X (CBSE)</i>	<i>85.66%</i>

CERTIFICATIONS

- Classify Images with TensorFlow on Google Cloud – Google (2025)
- DSA in Python – NPTEL (2025)
- Postman API Student Expert – Postman (2025)
- Master Machine Learning for Data Science & AI – Codebasics (2026)